# Define direction cosines and direction ratios of a straight line.

Direction cosines :- If a given line AB makes angles $\alpha, \beta, \gamma$ with axes of X, Y and Z respectively, then $\cos\alpha$, $\cos\beta$ and $\cos\gamma$ are called the direction cosines of the line AB and are generally denoted by the letters $l, m$ and $n$ respectively i.e. $l = \cos\alpha$, $m = \cos\beta$ and $n = \cos\gamma$.

Direction ratios :- Consider a line AB. Then any three numbers $a, b$ and $c$ which are proportional to the direction cosines $l, m, n$ respectively of the given line are called direction ratios of the given line.

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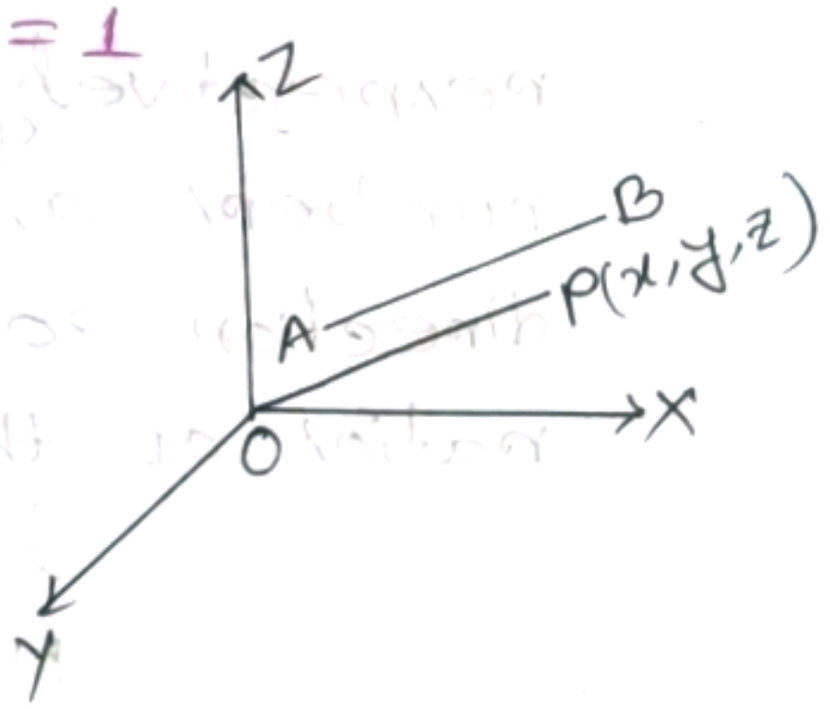
# Find the relation between direction cosines ##

If $l, m$ and $n$ be the direction cosines of a straight line, then show that

$$l^2 + m^2 + n^2 = 1$$

or $$\cos^2\alpha + \cos^2\beta + \cos^2\gamma = 1$$

Solution :- Let AB be the line whose direction cosines are $l, m, n$. Through the origin O draw a line OP parallel to AB and of length $r$ so that the



direction cosines of the line OP are also $l, m, n$.
Now if $(x, y, z)$ be the co-ordinates of the point P then we know that

$$x = lr \;,\; y = mr \;,\; z = nr$$

Squaring and adding,

$$x^2 + y^2 + z^2 = l^2 r^2 + m^2 r^2 + n^2 r^2$$
$$= r^2 (l^2 + m^2 + n^2) \quad - - - (1)$$

But $x^2 + y^2 + z^2 = OP^2 = r^2$ [length]

Hence from (1), we get

$$r^2 = r^2 (l^2 + m^2 + n^2)$$
$$\Rightarrow l^2 + m^2 + n^2 = 1$$
$$\Rightarrow \cos^2\alpha + \cos^2\beta + \cos^2\gamma = 1$$

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# What is the relation between direction cosines and direction ratios?

Solution :- Let us consider a line AB whose direction cosines are $\alpha, \beta, \gamma$ with axes of $x, y, z$ respectively. Again we know that any three numbers $a, b, c$ which are proportional to the direction cosines $l, m, n$ are called direction ratios of the given line. Therefore we get

$$l \propto a \text{ or } l = ka \text{ or } k = l/a$$
$$m \propto b \text{ or } m = kb \text{ or } k = m/b$$
$$n \propto c \text{ or } n = kc \text{ or } k = n/c$$

Now

$$l^2+m^2+n^2 = k^2a^2+k^2b^2+k^2c^2$$

$$\Rightarrow l^2+m^2+n^2 = k^2(a^2+b^2+c^2)$$

$$\Rightarrow k^2 = \frac{l^2+m^2+n^2}{a^2+b^2+c^2}$$

$$\Rightarrow k = \pm\sqrt{\frac{l^2+m^2+n^2}{a^2+b^2+c^2}}$$

$$= \pm\frac{1}{\sqrt{a^2+b^2+c^2}}$$

Then $\frac{l}{a} = \frac{m}{b} = \frac{n}{c} = \pm\frac{1}{\sqrt{a^2+b^2+c^2}}$

Thus $l = \frac{\pm a}{\sqrt{a^2+b^2+c^2}}$

$$m = \frac{\pm b}{\sqrt{a^2+b^2+c^2}}$$

$$n = \frac{\pm c}{\sqrt{a^2+b^2+c^2}}$$

which is the required relation.

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→ Projection of a line on another line

$$A'B' \text{ (Projection)} = AB\cos\theta$$

# The projection of a line on the co-ordinate axes are 2, 3 and 6. Find the length and the direction cosines of the line.

Solution:- Let the line be PQ and its direction cosines be $l, m, n$ then we know that

$$P'Q' = PQ\cos\theta \quad - - - (1)$$

where $P'Q'$ is the projection of PQ on a line AB and $\theta$ is the angle between PQ and AB. Now choosing AB as X-axis, Y-axis and Z-axis successively, we get

$$2 = PQ\cos\alpha \;,\; 3 = PQ\cos\beta \;,\; 6 = PQ\cos\gamma \quad ---- (2)$$

Squaring and adding,

$$4+9+36 = (PQ)^2(\cos^2\alpha + \cos^2\beta + \cos^2\gamma)$$

$$\Rightarrow (PQ)^2 = 49$$

$$\Rightarrow PQ = 7$$

which is the required length.

Putting this value in (2) the direction cosines are

$$\cos\alpha = \frac{2}{7} \;,\; \cos\beta = \frac{3}{7} \;,\; \cos\gamma = \frac{6}{7}$$

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# The projection of a line on the co-ordinate axes are 12, 4, 3 Find the length of the line and its direction cosines.

Solution:- Do yourself : Ans :- 13, $\frac{12}{13}$, $\frac{4}{13}$, $\frac{3}{13}$.

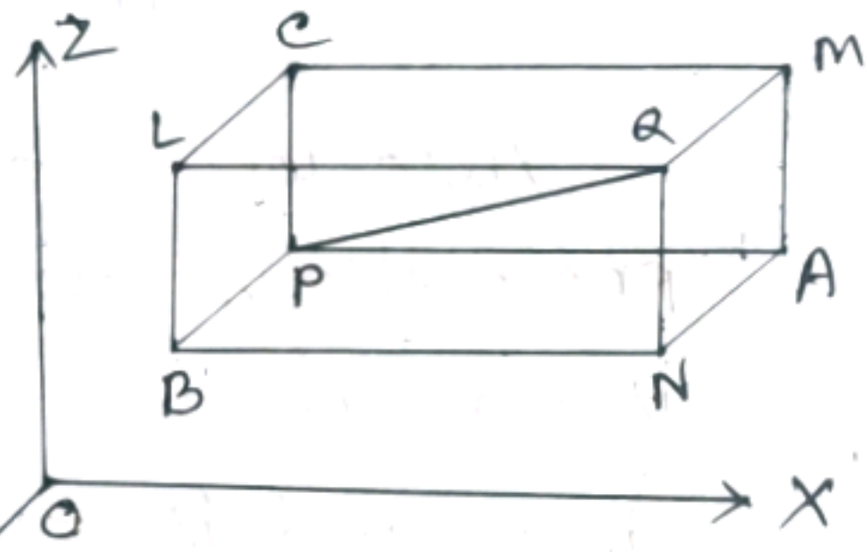
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# Prove that, the projection of a line joining $P(x_1, y_1, z_1)$ and $Q(x_2, y_2, z_2)$ on another line whose direction cosines are $l, m, n$ is

$$(x_2 - x_1)\,l + (y_2 - y_1)\,m + (z_2 - z_1)\,n$$

Solution:- Given points

$P(x_1, y_1, z_1)$, $Q(x_2, y_2, z_2)$.




Through P and Q draw planes parallel to the co-ordinate planes to form a rectangular parallelopiped. Then clearly

$$PA = x_2 - x_1, \quad NA = y_2 - y_1 \quad NQ = z_2 - z_1$$

If $\alpha, \beta, \gamma$ be the angles which the line whose direction cosines are $l, m, n$ makes with the axis, then

$$l = \cos\alpha, \quad m = \cos\beta, \quad n = \cos\gamma$$

Now the projection of PA which is parallel to X-axis on that line is

$$PA\cos\alpha = (x_2 - x_1)\,l$$

Similarly,

the projection of AN which is parallel to Y-axis on that line is

$$AN\cos\beta = (y_2 - y_1)\,m$$

and the projection of NQ which is parallel to Z-axis on that line is

$$NQ\cos\gamma = (z_2 - z_1)\,n$$

Also we know that the sum of the projections of PA, AN and NQ on a line is equal to the projection of PQ on that line and is therefore equal to

$$(x_2 - x_1)\,l + (y_2 - y_1)\,m + (z_2 - z_1)\,n.$$

(Proved)

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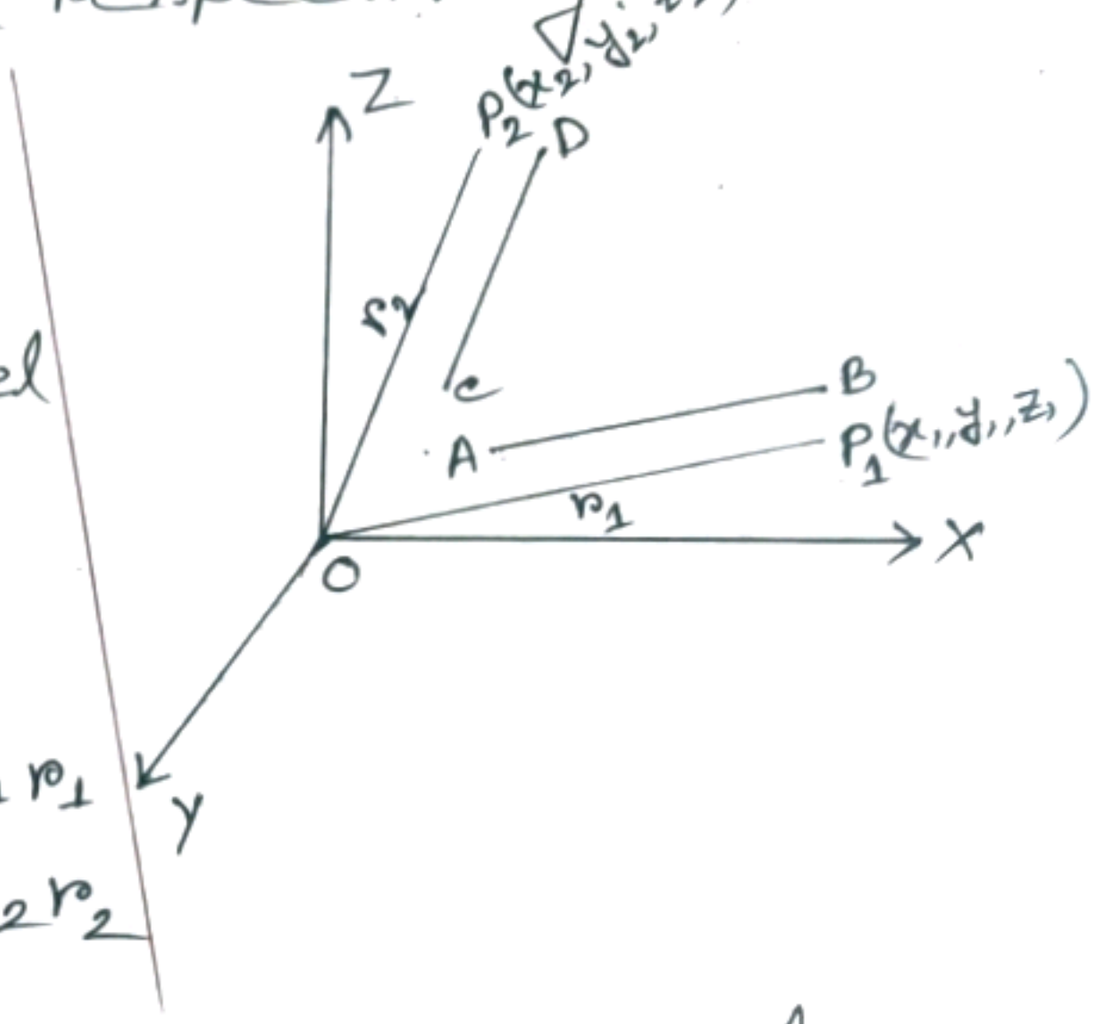
# 1(b) →

\# If $l_1, m_1, n_1$ and $l_2, m_2, n_2$ be the direction cosines of any two lines AB and CD and $\theta$ be the angle between them, then

$$\cos\theta = l_1 l_2 + m_1 m_2 + n_1 n_2$$

Solution:- Given lines AB and CD with direction cosines $l_1, m_1, n_1$ and $l_2, m_2, n_2$ respectively.

Through the origin O draw two lines $OP_1$ of length $r_1$ and $OP_2$ of length $r_2$ parallel respectively to the lines AB and CD. Then




$$x_1 = l_1 r_1, \quad y_1 = m_1 r_1, \quad z_1 = n_1 r_1$$

and $x_2 = l_2 r_2 \quad y_2 = m_2 r_2 \quad z_2 = n_2 r_2$

If $\theta$ be the angle between the given lines, then $\theta$ is also equal to the angle between $OP_1$ and $OP_2$.

Now the projection of line $OP_2$ joining $(0,0,0)$ and $(x_2, y_2, z_2)$ on the line $OP_1$ whose directions cosines are $(l_1, m_1, n_1)$ is

$$(x_2 - 0) l_1 + (y_2 - 0) m_1 + (z_2 - 0) n_1$$

$$= l_1 x_2 + m_1 y_2 + n_1 z_2$$

But the projection of $OP_2$ on $OP_1$ is

$$OP_2 \cos\theta$$

Therefore,

$$OP_2 \cos\theta = l_1 x_2 + m_1 y_2 + n_1 z_2$$

$$\text{or } r_2 \cos\theta = l_1 (l_2 r_2) + m_1 (m_2 r_2) + n_1 (n_2 r_2)$$

$$\therefore \cos\theta = l_1 l_2 + m_1 m_2 + n_1 n_2$$

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**Remember:-** If $P(x_1, y_1, z_1)$ and $Q(x_2, y_2, z_2)$ be any two points then

① the direction ratios of the line PQ are

$$x_2 - x_1 , \; y_2 - y_1 , \; z_2 - z_1$$

② Consider a line AB with direction cosines $l_1, m_1, n_1$ and a line CD with " " $l_2, m_2, n_2$ and $\theta$ be the angle between them then $\cos\theta = l_1 l_2 + m_1 m_2 + n_1 n_2$

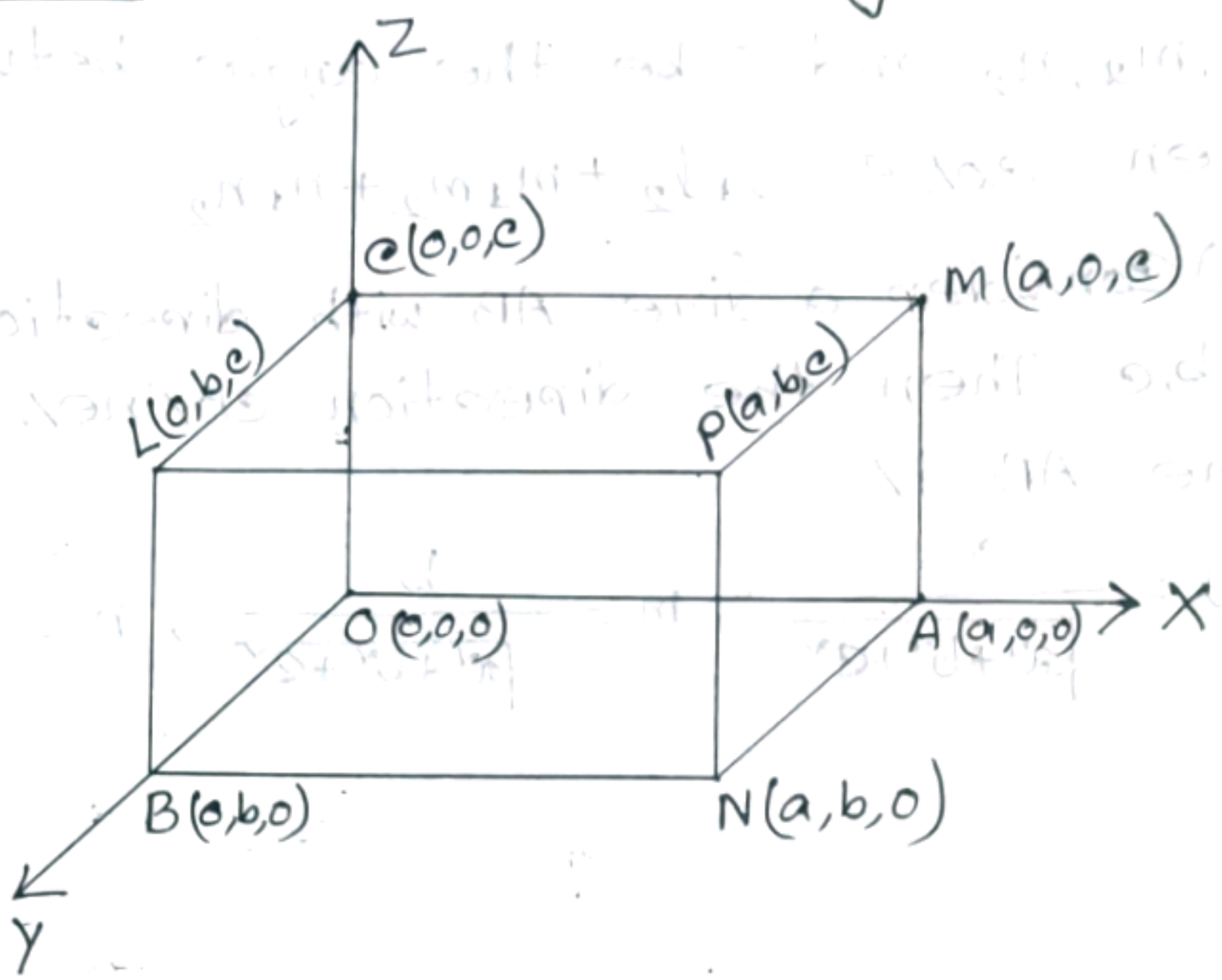
③ Consider a line AB with direction ratios a, b, c. Then the direction cosines of the line AB is

$$l = \frac{a}{\sqrt{a^2+b^2+c^2}} , \; m = \frac{b}{\sqrt{a^2+b^2+c^2}} , \; n = \frac{c}{\sqrt{a^2+b^2+c^2}}$$

# If the edges of a rectangular parallelopiped be a, b, c show that the angles between the four diagonals are given by

$$\cos^{-1}\frac{\pm a^2 \pm b^2 \pm c^2}{a^2+b^2+c^2}$$

Solution :- Consider a rectangular parallelopiped




The four diagonals are AL, BM, CN and OP where the coordinates of O are (0,0,0) P(a,b,c), A(a,0,0), B(0,b,0), C(0,0,c), L(0,b,c), M(a,0,c) and N(a,b,0).

Direction ratios of OP are

a-0, b-0, c-0

i.e. a, b, c

And direction cosines are

$$\frac{a}{\sqrt{\Sigma a^2}}, \frac{b}{\sqrt{\Sigma a^2}}, \frac{c}{\sqrt{\Sigma a^2}}$$

$[a^2+b^2+c^2 = \Sigma a^2]$ —

Again, direction ratios of AL' are

$$0-a, b-0, c-0$$

i.e. $-a, b, c$

and direction cosines are

$$\frac{-a}{\sqrt{\Sigma a^2}}, \frac{b}{\sqrt{\Sigma a^2}}, \frac{c}{\sqrt{\Sigma a^2}}$$

Similarly we can find the direction cosines and ratios of the other lines.

If $\theta$ be the angle between OP and AL, we get

$$\cos\theta = \frac{a}{\sqrt{\Sigma a^2}} \cdot \frac{-a}{\sqrt{\Sigma a^2}} + \frac{b}{\sqrt{\Sigma a^2}} \cdot \frac{b}{\sqrt{\Sigma a^2}} + \frac{c}{\sqrt{\Sigma a^2}} \cdot \frac{c}{\sqrt{\Sigma a^2}}$$

$$= \frac{-a^2+b^2+c^2}{a^2+b^2+c^2}$$

Similarly we can find the angle between other pairs and hence the angles in general between the six pairs are given by

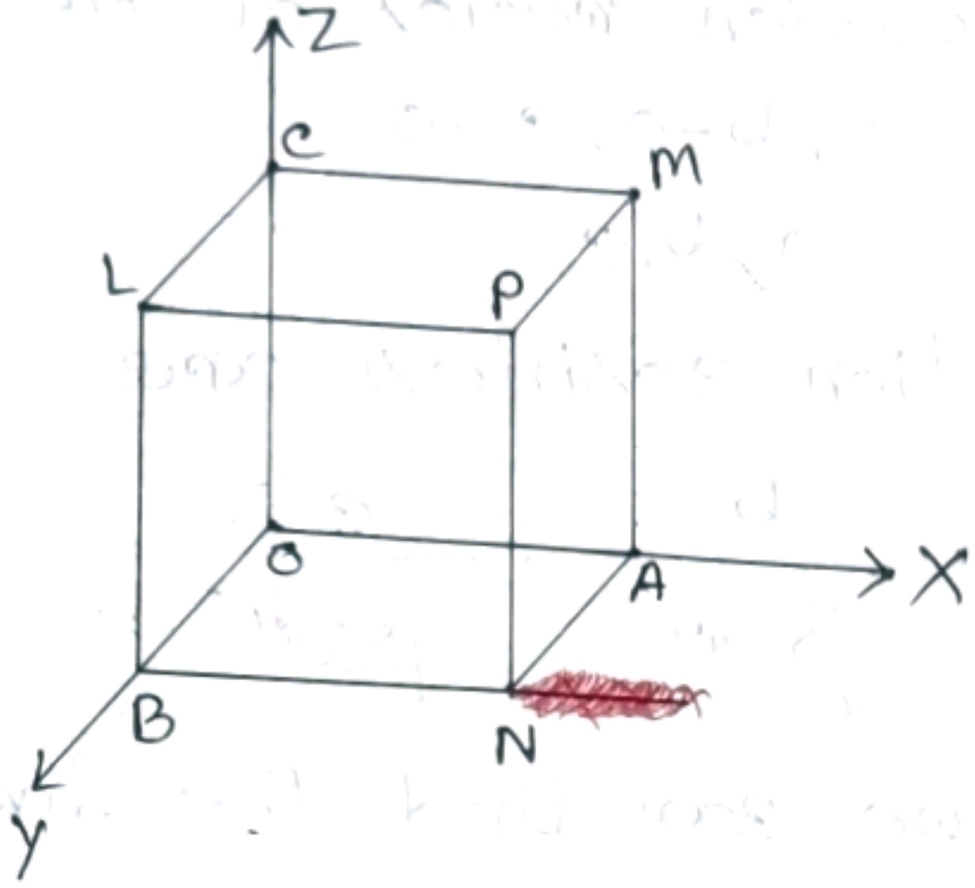
$$\theta = \cos^{-1} \frac{\pm a^2 \pm b^2 \pm c^2}{a^2+b^2+c^2}$$

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# A line makes angles $\alpha, \beta, \gamma, \delta$ with the four diagonals of a cube. Show that

$$\cos^2\alpha + \cos^2\beta + \cos^2\gamma + \cos^2\delta = \frac{4}{3}$$

**Solution:-** ~~Consider a cube~~




The direction cosines of the four diagonals of a cube on putting $a=b=c$ are

$$OP\left(\frac{1}{\sqrt{3}}, \frac{1}{\sqrt{3}}, \frac{1}{\sqrt{3}}\right), \quad AL\left(\frac{-1}{\sqrt{3}}, \frac{1}{\sqrt{3}}, \frac{1}{\sqrt{3}}\right)$$

$$BM\left(\frac{1}{\sqrt{3}}, \frac{-1}{\sqrt{3}}, \frac{1}{\sqrt{3}}\right), \quad CN\left(\frac{1}{\sqrt{3}}, \frac{1}{\sqrt{3}}, \frac{-1}{\sqrt{3}}\right)$$

Let $l, m, n$ be direction cosines of a line which is inclined at angles $\alpha, \beta, \gamma, \delta$ respectively to the four diagonals, then

$$\cos\alpha = l \cdot \frac{1}{\sqrt{3}} + m \cdot \frac{1}{\sqrt{3}} + n \cdot \frac{1}{\sqrt{3}}$$

$$= \frac{l+m+n}{\sqrt{3}}$$

Similarly,

$$\cos\beta = \frac{-l+m+n}{\sqrt{3}}$$

$$\cos\gamma = \frac{l-m+n}{\sqrt{3}}$$

$$\cos\delta = \frac{l+m-n}{\sqrt{3}}$$

Therefore,

$$\cos^2\alpha + \cos^2\beta + \cos^2\gamma + \cos^2\delta$$

$$= \frac{1}{3}\left[(l+m+n)^2 + (-l+m+n)^2 + (l-m+n)^2 + (l+m-n)^2\right]$$

$$= \frac{1}{3} \cdot 4\left(l^2+m^2+n^2\right)$$

$$= \frac{4}{3} \quad [\because l^2+m^2+n^2 = 1]$$

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Consider a cube whoose every side is a

Fig

The four diagonals are OP, AL, BM, CN where the co-ordinates of

O P A L B M C N

Now direction ratios of OP are

a-0, a-0, a-0

i.e. a, a, a

and direction cosines of OP are

$$\frac{a}{\sqrt{a^2+a^2+a^2}}$$

i.e. $\frac{1}{\sqrt{3}}, \frac{1}{\sqrt{3}}, \frac{1}{\sqrt{3}}$

Similarly for the diagonals AL, BM, CN the direction cosines are

AL $\left(\frac{-1}{\sqrt{3}}, \frac{1}{\sqrt{3}}, \frac{1}{\sqrt{3}}\right)$ BM $\left(\frac{1}{\sqrt{3}}, \frac{-1}{\sqrt{3}}, \frac{1}{\sqrt{3}}\right)$ CN $\left(\frac{1}{\sqrt{3}}, \frac{1}{\sqrt{3}}, \frac{-1}{\sqrt{3}}\right)$

Let l, m, n, . . . ——— sheet